

May 31, 2026

The Editor
Communications Earth & Environment

Dear Editor,

We are pleased to submit our manuscript, “Reproducible model-based AI planning for county-scale farmland consolidation in fragmented mountain landscapes,” for consideration as a Research Article in *Communications Earth & Environment*.

Farmland consolidation in mountainous terrain is one of the largest spatial-planning levers for food security, slope erosion control, and rural land governance, yet county-scale optimisation has remained operationally inaccessible to routine planning offices: reinforcement-learning baselines on this task require 8–12 GPU-hours per random seed with non-trivial seed collapse, and classical operations-research methods at matched 180s wall-clock budgets reach less than 11% of our pipeline’s reward because the simulator’s per-step cost (17.9 ms/step) limits their iteration budget below the convergence threshold.

The manuscript reports a learned-surrogate model-predictive-control pipeline that delivers a slope reduction of $-1.289 \pm 0.079\%$ on Bishan District (Chongqing, 52,515 parcels), $1.6\times$ the magnitude reached by in-house centralised-PPO and multi-agent baselines, while reducing wall time by roughly two orders of magnitude (8–12 GPU-hours per seed reduced to ~ 7 minutes per scenario on a 12-thread desktop CPU). The result reproduces on Neijiang Dongxing (Sichuan, 76,376 parcels), and underlying gains are validated by independent GIS recomputation directly from the optimised cadastral shapefile.

Four aspects align with the journal’s mission. The work is an actionable Earth & environmental science contribution, packaged as both an ArcGIS Pro Python Toolbox and a non-commercial QGIS-compatible CLI. It is fully reproducible: trained ensembles (~ 60 checkpoints), a deterministic seven-preset synthetic benchmark under CC-BY 4.0, a complete public-data Buchanan VA mine-restoration cross-domain case (entirely US-government open data), and a reproduction recipe are publicly available now (not gated on publication) at github.com/zhouning/arcgis-farmland-mpc. The manuscript treats restricted data transparently: the raw cadastral records for the two Chinese counties cannot be publicly redistributed under data-governance restrictions, but the synthetic benchmark and Buchanan restoration case jointly form a fully open reproduction track requiring no restricted-data access. We close with a falsifiable problem-selection criterion: the pipeline’s advantage over classical operations-research baselines holds when simulator step cost exceeds ~ 30 ms or when reward exhibits combinatorial cost interactions, giving practitioners a concrete two-quantity test applicable to hydrologic-model-driven watershed restoration, integrated power-grid maintenance scheduling, and other Earth-science problems with expensive simulators.

The manuscript has not been submitted elsewhere. The authors have no competing interests.

Sincerely,

Ning Zhou
Public R&D Center, SuperMap
Software Co., Ltd.
Beijing, China
zhouning1@supermap.com